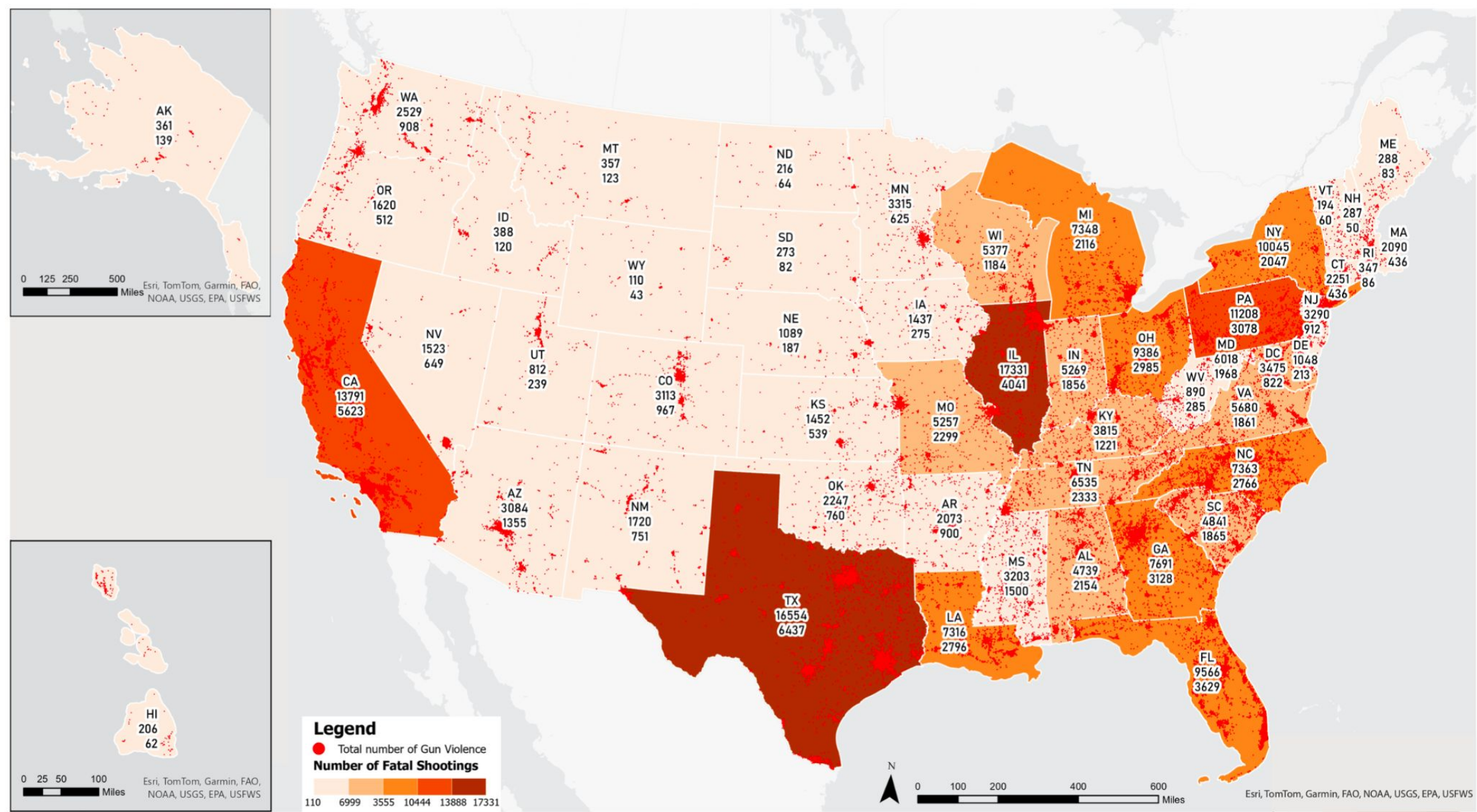


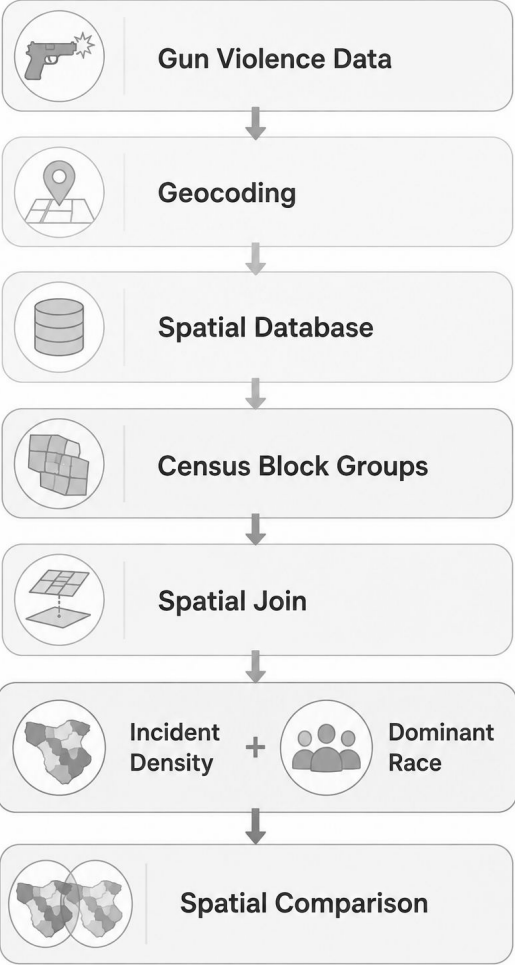
SPATIAL ANALYSIS OF GUN VIOLENCE & NEIGHBORHOOD DEMOGRAPHICS

A multi-scale GIS investigation of spatial disparities in fatal shootings across the United States

Total Gun Violence in United States



How do neighborhood demographic patterns relate to the geographic distribution of fatal shootings across different U.S. cities?

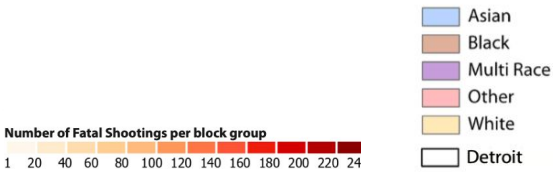


Spatial Data Integration

National-scale mapping reveals broad regional patterns, while neighborhood-scale analysis exposes localized hotspots that become invisible at larger geographic scales.

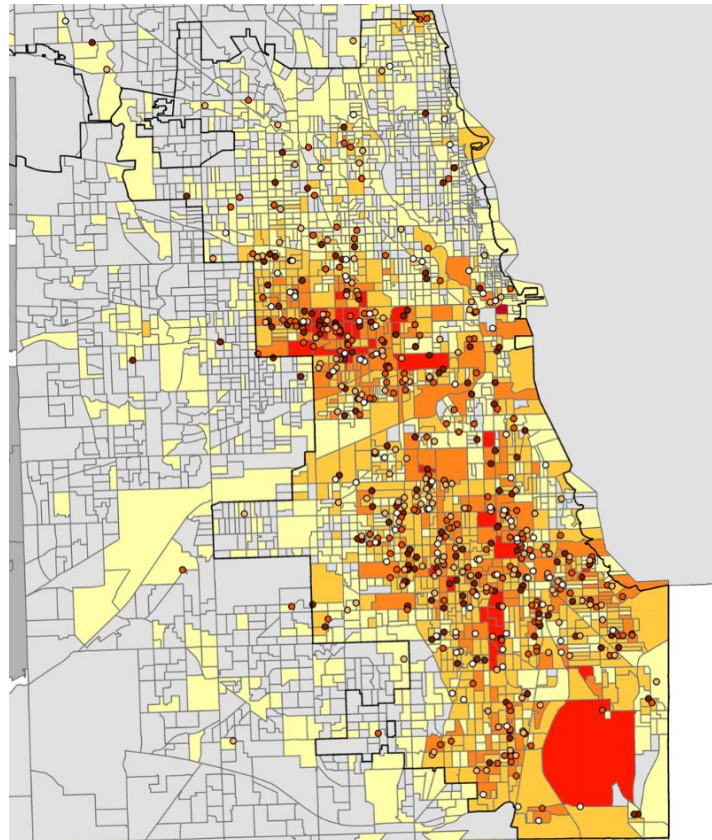
NEIGHBORHOOD-LEVEL SPATIAL ANALYSIS

Comparing fatal shooting density with dominant neighborhood demographics across four U.S. cities.

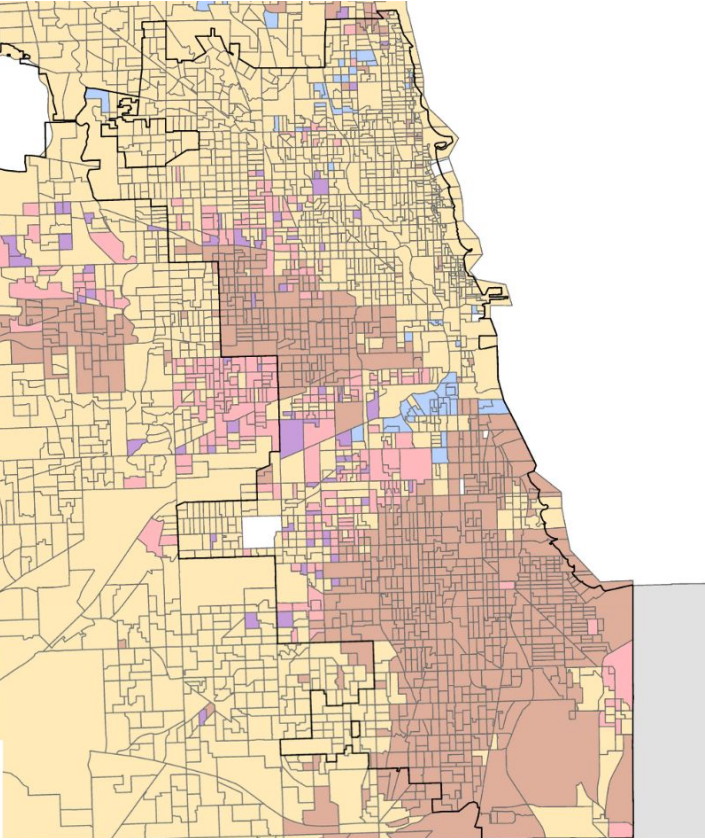


CHICAGO , IL

Gun Violence



Neighborhood Demographics



Spatial Pattern

Distinct hotspot clusters emerge across Chicago's South and West Side neighborhoods, revealing concentrated fatal shootings at the Census Block Group level.

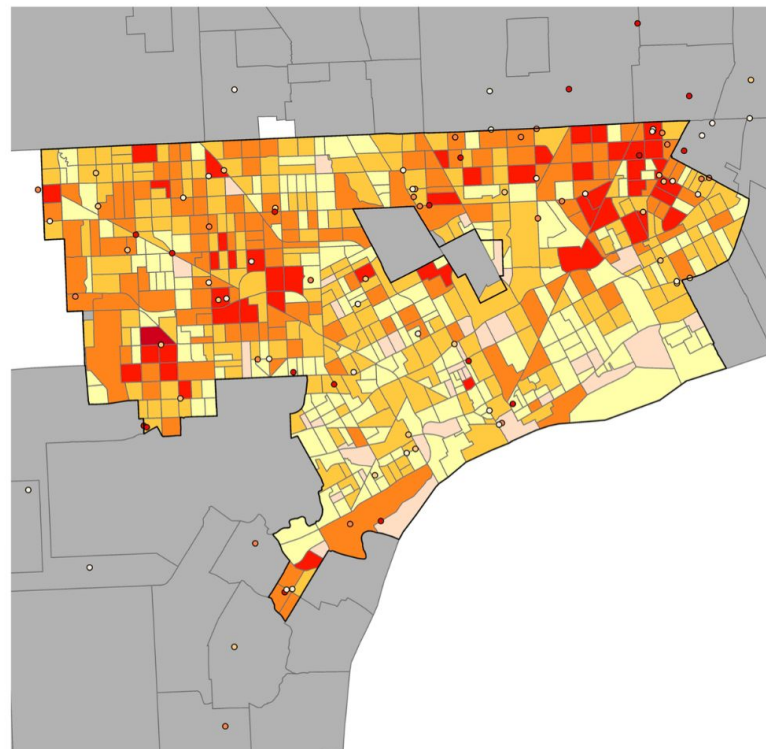
Insight

Localized hotspot clusters demonstrate the value of neighborhood-scale GIS, revealing spatial variation that is often concealed in citywide analyses.

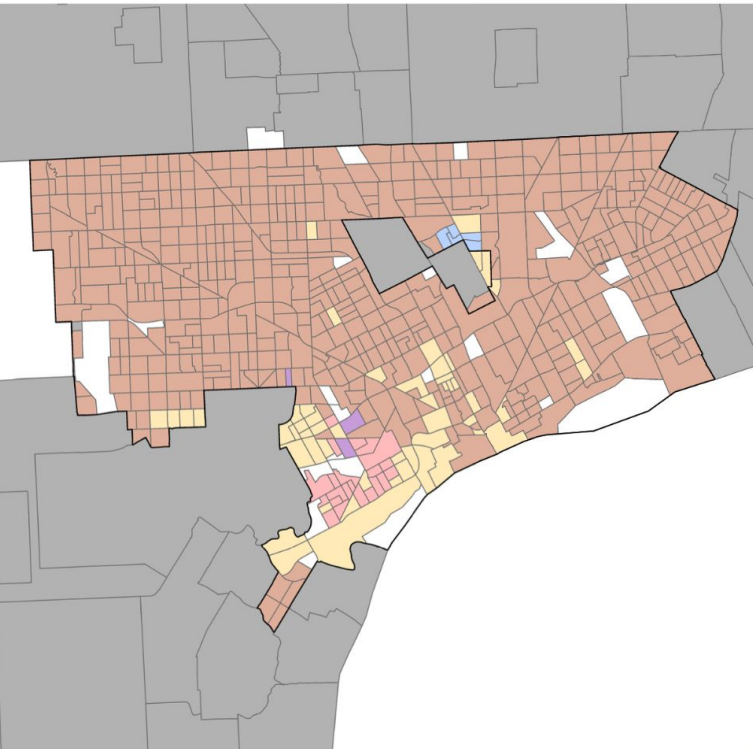


DETROIT , MI

Gun Violence



Neighborhood Demographics



Spatial Pattern

Detroit exhibits the strongest and most continuous hotspot clusters among the study cities, with fatal shootings concentrated across central neighborhoods.

Insight

The continuous clustering of incidents suggests persistent spatial concentration, making Detroit a strong example of how GIS reveals place-based inequalities.

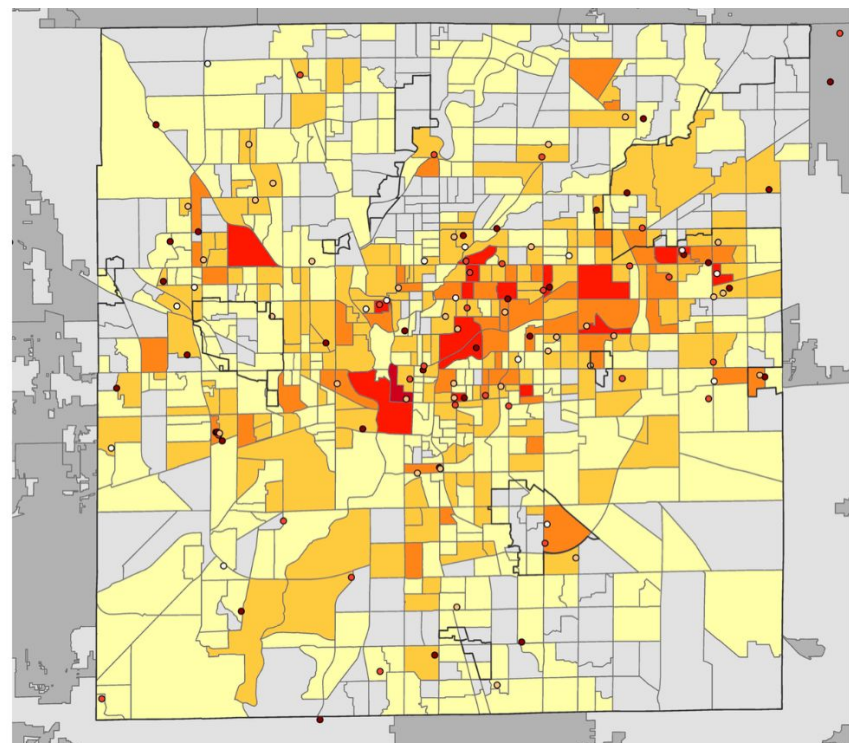


NEIGHBORHOOD-LEVEL SPATIAL ANALYSIS

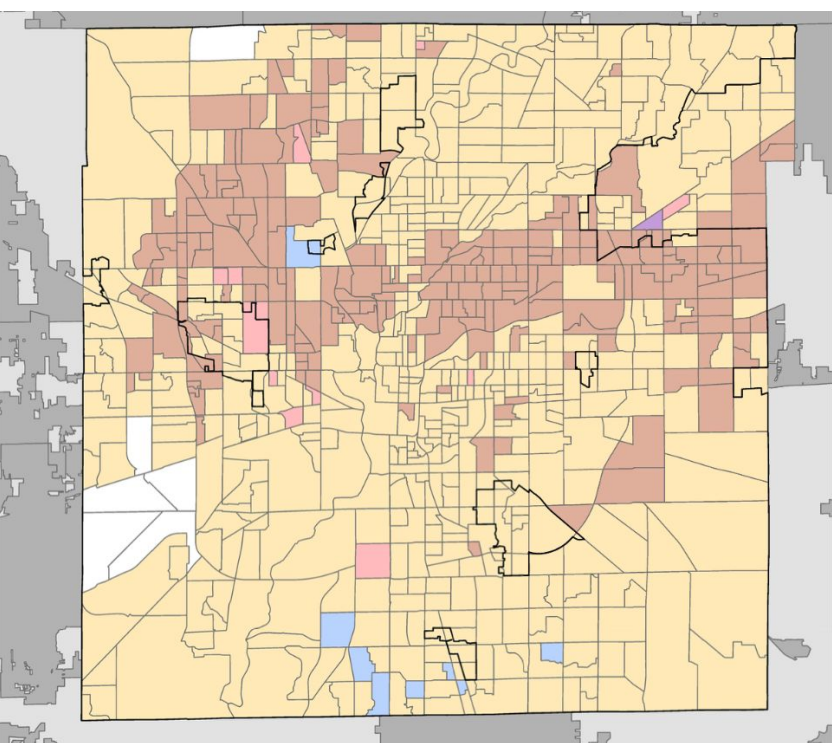
Comparing fatal shooting density with dominant neighborhood demographics across four U.S. cities.

INDIANAPOLIS , IN

Gun Violence



Neighborhood Demographics



Spatial Pattern

Fatal shootings are concentrated within Indianapolis' urban core, with distinct hotspot clusters surrounded by lower-density neighborhoods.

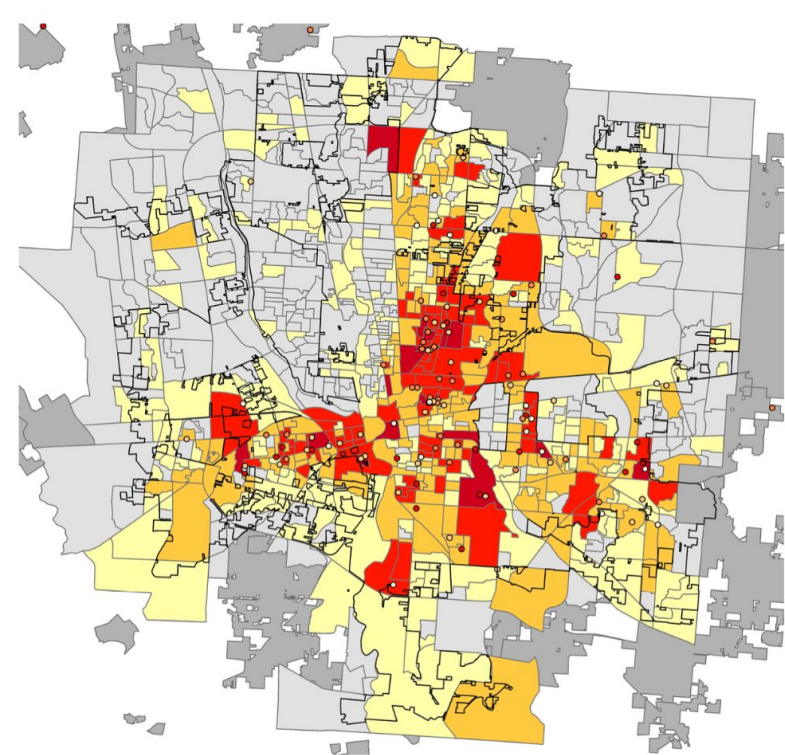
Insight

Comparing incident density with neighborhood demographics reveals localized spatial disparities that support neighborhood-scale planning and targeted community investment.

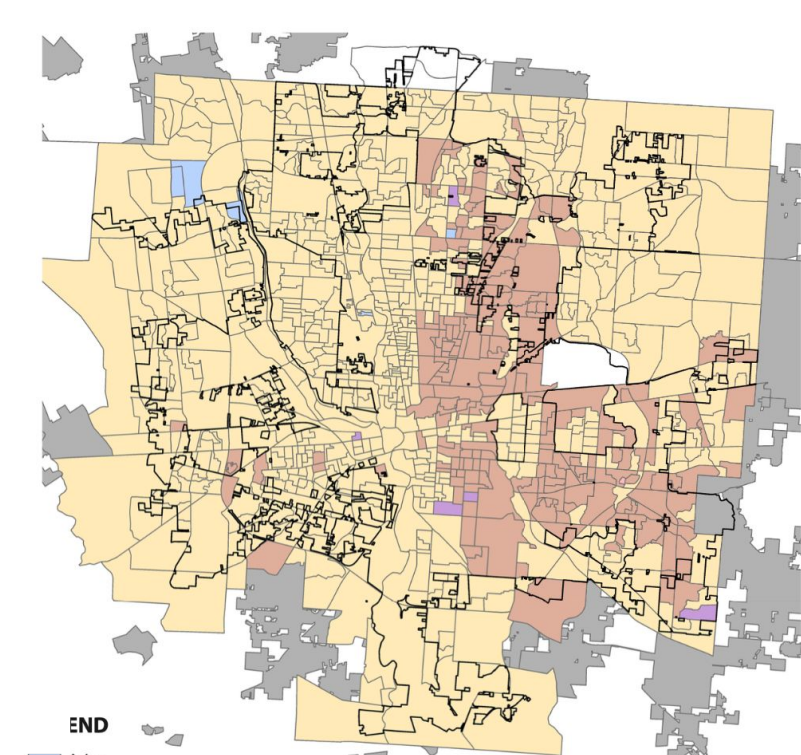


COLUMBUS , OH

Gun Violence



Neighborhood Demographics



Spatial Pattern

Columbus exhibits several localized hotspot clusters concentrated around the central city, while surrounding neighborhoods experience comparatively lower incident density.

Insight

The spatial comparison demonstrates how GIS uncovers neighborhood-level variation, providing valuable context for understanding geographic disparities across the city.

